

How the chooser decides

Geometry chooses the action. Pose matching chooses its animation.

The runtime check supplies ledge shape, obstacle height and depth, speed, movement mode and pose history. The top-level table selects Hurdle, Vault or Mantle; its nested table filters available clips. Pose Search compares passing rows and returns a montage and start time.

01 / GEOMETRY	02 / ANIMATION
Front ledge, back ledge, back floor, depth and back drop select an action family.	Height, depth, speed and ground/air state filter the family's rows. Pose history selects among matches.

Top-level rules

Result	Front	Back ledge	Back floor	Depth	Back drop
Hurdle	true	true	true	0 - 59	50 - +
Mantle	true	true	true	0 - 59	0 - 10
Vault	true	true	false	0 - 59	any
Mantle	true	any	any	59 - +	any

Read the result correctly

Height does not turn a vault into a mantle. An action family can be geometrically valid yet have no suitable animation. A failed traversal falls back to the pawn's jump. Landing room, headroom, reach and body clearance must also pass.

Units: geometry ranges are centimeters; speed ranges are centimeters per second; sampling ranges are montage seconds. Boundaries follow Unreal's chooser range behavior. 'Any' means that column does not filter this row.

This reference describes the current generated Neutral traversal chooser: a 300 cm grounded climb ceiling and running-climb samples through 0.60 seconds. It is regenerated from the shipping manifest. An older generated chooser needs its setup-row rebuild; a custom chooser is not silently replaced.

Hurdles / animation rows

Each row must pass every filter before pose matching compares it.

Clip suffix	Height	Depth	Speed	Mode	Sample
Hurdle_1_0_stand_F_V2_Lfoot	0 - 125	0 - 25	0 - 100	OnGround	0 - 0.07
Hurdle_1_0_walk_F_Lfoot	0 - 125	0 - 25	100 - 250	OnGround	0 - 0.73
Hurdle_1_0_walk_F_Rfoot	0 - 125	0 - 25	100 - 250	OnGround	0 - 0.84
Hurdle_1_0_run_F_Lfoot	0 - 125	0 - 25	250 - +	OnGround	0 - 0.49
Hurdle_1_0_run_F_Rfoot	0 - 125	0 - 25	250 - +	OnGround	0 - 0.48
Hurdle_1_0_stand_F_V2_Rfoot	0 - 125	25 - 60	0 - 100	OnGround	0 - 0.12
Hurdle_1_0_walk_F_V2_Lfoot	0 - 125	25 - 60	100 - 250	OnGround	0 - 0.5
Hurdle_1_0_walk_F_V2_Rfoot	0 - 125	25 - 60	100 - 250	OnGround	0 - 0.5
Hurdle_1_0_run_F_V2_Lfoot	0 - 125	25 - 60	25 - +	OnGround	0 - 0.66
Hurdle_1_0_run_F_V2_Rfoot	0 - 125	25 - 60	25 - +	OnGround	0 - 0.67
Catch_Hurdle_low_stand	0 - 50	0 - 60	any	InAir	0.83 - 0.87
Catch_Hurdle_med_stand	50 - 100	0 - 60	any	InAir	0.83 - 0.87
Catch_Hurdle_high_stand	100 - 200	0 - 60	any	InAir	0.88 - 0.91

Clip suffix key

Unprefixed names use **AM_M_Neutral_Traversal_**. Names marked Relaxed use **AM_M_Relaxed_Traversal_**. Sampling limits constrain pose matching's chosen start; they do not trim or replace the animation.

Pose Search database: **PSD_Hurdles_0**. The generated database and schema belong to the selected skeleton; animations for another skeleton are not interchangeable.

Mantles / animation rows

Each row must pass every filter before pose matching compares it.

Clip suffix	Height	Depth	Speed	Mode	Sample
Mantle_1_0_stand_F_Lfoot	0 - 150	any	0 - 100	OnGround	0 - 0.1
Mantle_1_0_stand_F_Rfoot	0 - 150	any	0 - 100	OnGround	0 - 0.1
Mantle_1_0_walk_F_Lfoot	0 - 150	any	100 - 250	OnGround	0 - 0.37
Mantle_1_0_walk_F_Rfoot	0 - 150	any	100 - 250	OnGround	0 - 0.75
Mantle_1_0_run_F_Lfoot	0 - 150	any	250 - +	OnGround	0 - 0.33
Mantle_1_0_run_F_Rfoot	0 - 150	any	250 - +	OnGround	0 - 0.32
Climb_Start_2_5_stand_F_Lfoot	150 - 300	any	0 - 100	OnGround	0 - 0.17
Climb_Start_2_5_stand_F_Rfoot	150 - 300	any	0 - 100	OnGround	0 - 0.16
Climb_Start_2_5_walk_F_Lfoot	150 - 300	any	100 - 250	OnGround	0 - 0.95
Climb_Start_2_5_walk_F_Rfoot	150 - 300	any	100 - 250	OnGround	0 - 1.03
Climb_Start_2_5_run_F_Lfoot	150 - 300	any	250 - +	OnGround	0 - 0.6
Climb_Start_2_5_run_F_Rfoot	150 - 300	any	250 - +	OnGround	0 - 0.6
Catch_Mantle_low_stand	0 - 50	any	any	InAir	0.87 - 0.9
Catch_Mantle_med_stand	50 - 100	any	any	InAir	0.84 - 0.87
Catch_Cliff_high_standF	100 - 200	any	any	InAir	0.91 - 0.94

Clip suffix key

Unprefixed names use **AM_M_Neutral_Traversal_**. Names marked Relaxed use **AM_M_Relaxed_Traversal_**. Sampling limits constrain pose matching's chosen start; they do not trim or replace the animation.

Pose Search database: **PSD_Mantles_0**. The generated database and schema belong to the selected skeleton; animations for another skeleton are not interchangeable.

Vaults / animation rows

Each row must pass every filter before pose matching compares it.

Clip suffix	Height	Depth	Speed	Mode	Sample
Vault_1_0_stand_F_Lfoot	0 - 125	0 - 60	0 - 100	OnGround	0 - 0.08
Vault_1_0_walk_F_Rfoot	0 - 125	0 - 60	100 - 250	OnGround	0 - 0.51
Vault_1_0_run_F_Lfoot	0 - 125	0 - 60	250 - +	OnGround	0 - 0.61
Catch_Vault_low	0 - 50	0 - 60	any	InAir	0.88 - 0.91
Catch_Vault_med	50 - 80	0 - 60	any	InAir	0.87 - 0.9
Catch_Vault_high	80 - 200	0 - 60	any	InAir	0.85 - 0.89

Clip suffix key

Unprefixed names use **AM_M_Neutral_Traversal_**. Names marked Relaxed use **AM_M_Relaxed_Traversal_**. Sampling limits constrain pose matching's chosen start; they do not trim or replace the animation.

Pose Search database: **PSD_Vaults_0**. The generated database and schema belong to the selected skeleton; animations for another skeleton are not interchangeable.

Height coverage and troubleshooting

Grounded family	Available height bands, cm
Hurdles	0 - 125
Mantles	0 - 150; 150 - 300
Vaults	0 - 125

A 300 cm ceiling is an animation limit, not a promise of climbability. Standard containers still need usable roof depth, valid landing space and local proxies that follow their actual ledge. Inspect approach sides separately.

Symptom	Inspect
Wrong action family	Back ledge, back floor, obstacle depth and back drop. Start with the top-level rules.
Valid geometry, no montage	The chosen family's height/depth/speed/mode filters and the selected skeleton's imported clips.
Old 275 cm cutoff	Rebuild the generated chooser with the current setup row. Confirm a custom table is not still assigned.
Another skeleton's animation	Use that skeleton's owned import and generated chooser. Preserve custom tables and backups.
Body clips during a successful climb	Inspect animation proportions and actual geometry. Successful selection and warp alignment alone do not prove clearance.

Back up before changing your setup

Keep a source-control checkpoint or complete project copy. Rebuilding a generated chooser keeps its previous chooser/schema as numbered backups. Importing and setup may also change shared skeletons, Blueprints and configuration.

Manifest SHA256: d3469600c92f92955ccc